

1. General Calibration Instructions

Calibration Gases:

Use only dry calibration gases.

Always apply a calibration gas pressure-free (i.e. via gas bag).

Apply calibration gas via probe.

Tightness check:

Apply span gas (O₂ ca. 2-3 vol %).

By drifts > 0,2 vol % check the total gas path on leakages.

Soot measurement:

Check or calibrate the soot measurement first after completion of the tightness check.

T-Gas:

By check / calibration in dip pod, release gas measuring tubing from instrument (otherwise undesirable heat removal thru measuring pump).

T-Air:

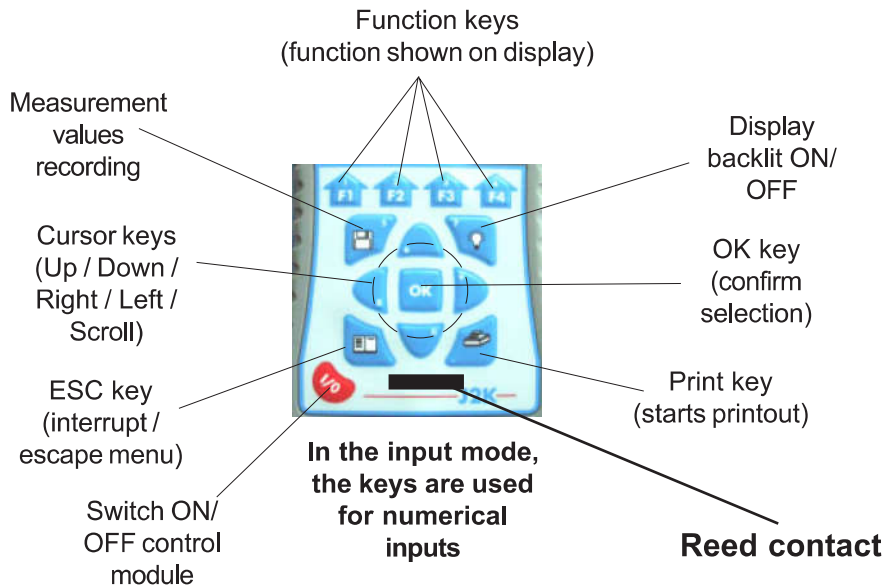
By check / calibration in liquid baths use non-conducting liquids to avoid false measurement results (after check clean the sensor with spirit)

Draught:

The draught sensor can only be calibrated within the low pressure range whereby the einzustellende draught must amount 0,5 hPa at a minimum.

12. ecom-J2K(N)

Notes



- Place control module in basic module or connect control & basic modules via cable.
- Switch on instrument (switch located below condensation trap).
- Start gas analysis ("Gas analysis" / <OK>).
- Confirm fuel type selection with "OK".
- Answer question about data processing with "NO" (<F4>).
- Wait for termination of calibration phase (1 minute).
- Switch to control menu (<ESC> / "Control" / <OK>).
- If need be, deactivate dongle.
- Activate reed contact with magnet (see drawing).
- Use arrow keys to select channel to be calibrated.
 - Calibrate CO / NO / NO2 / SO2.
- **(Important: Always apply span gas pressure-free!)**
 - Flow span gas.
 - Wait for stabilisation.
 - By drift press <OK>.
 - Input reference value.
 - Confirm input with <OK>.
- Calibrate T.Gas / T.Air.
 - Create reference temperature.
 - By drift press <OK>.
 - Input reference value.
 - Confirm input with <OK>.
- Calibrate draught.
 - Zero draught sensor with <F4>.
 - Create reference pressure.
 - By drift press <OK>.
 - input reference value.
 - Confirm input with <OK>.

- Calibrate soot measurement.
 - Connect appropriate reference system for volume determination at probe inner pipe.
 - Insert soot filter paper.
 - Press <OK>.
 - Start sucking process with <OK>.
 - Once 1.63 l has been sucked in, press again <OK> (pump operation time is recorded).
 - Press <ESC> to escape the calibration mode.

Notes